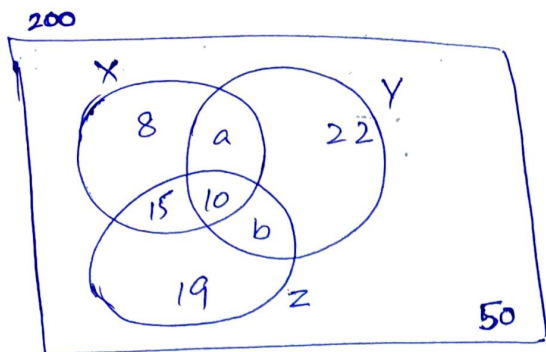


Q1:



10/10

(4)

$$X = \frac{1}{2}Z$$

$$8 + 15 + 10 + a = \frac{19 + 10 + 15 + b}{2}$$

$$33 + a = \frac{44 + b}{2}$$

$$33 + a = 22 + \frac{b}{2}$$

$$a - \frac{b}{2} = 22 - 33$$

$$a - \frac{b}{2} = -11 \quad \text{--- (i)}$$

$$8 + 15 + 10 + 19 + 22 + a + b + 50 = 200$$

$$124 + a + b = 200$$

$$a + b = 200 - 124$$

$$a + b = 76 \quad \text{--- (ii)}$$

(3)

$$a - \frac{b}{2} = -11$$

$$a + b = 76$$

$$+ (3b)/2 = + 87$$

$$3b = 87 \times 2 = 174$$

$$b = 174/3$$

$$b = 58$$

put in (ii)

$$a + b = 76$$

$$a + 58 = 76$$

$$a = 76 - 58$$

$$a = 18$$

(3)

(1)

Q2: 1. $(A \cup B) - C = (A - C) \cup (B - C)$

10/10

L.H.S

A	B	C	$(A \cup B)$	$(A \cup B) - C$
0	0	0	0	0
0	0	1	0	0
0	1	0	1	1
0	1	1	1	0
1	0	0	1	1
1	0	1	1	0
1	1	0	1	1
1	1	1	1	0

R.H.S

$(A - C) \cup (B - C)$

A	B	C	$A - C$	$B - C$	$(A - C) \cup (B - C)$
0	0	0	0	0	0
0	0	1	0	0	0
0	1	0	0	1	1
0	1	1	0	0	0
1	0	0	1	0	1
1	0	1	0	0	0
1	1	0	1	1	1
1	1	1	0	0	0

L.H.S = R.H.S proved.

2. ~~$(A \cap B)' - C = A' \cup B'$~~

$(A \cap B)' = A' \cup B'$

A	B	$A \cap B$	$(A \cap B)'$	A'	B'	$A' \cup B'$
0	0	0	1	1	1	1
0	1	0	1	1	0	1
1	0	0	1	0	1	1
1	1	1	0	0	0	0

L.H.S = R.H.S

Proved.

Q3: $(A-B)-C = (A-C)-B$

10/10

L.H.S = $(A-B)-C$

$(A \cap B') - C$

// Alternative Representation of set difference

$(A \cap B') \cap C'$

// set difference alternative

$A \cap (B' \cap C')$

// Associative Law

$A \cap (C' \cap B')$

// commutative Law

$(A \cap C') \cap B'$

// Associative Law

$(A-C) \cap B'$

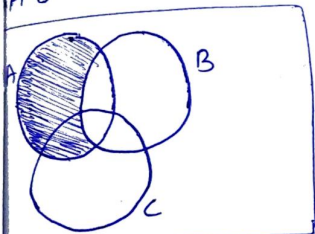
// set difference alternative

$(A-C)-B$

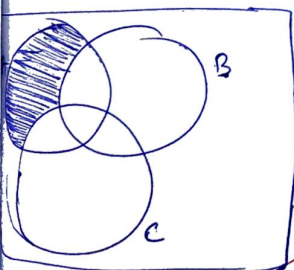
// set difference alternative

= R.H.S

L.H.S

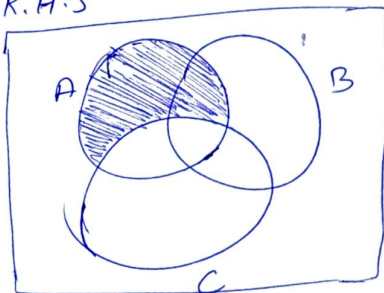


$A-B$

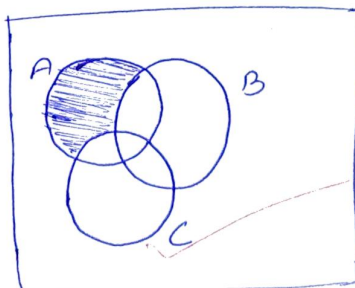


$(A-B)-C$

R.H.S



$A-C$



$(A-C)-B$

Both Venn Diagram are
Same L.H.S = R.H.S

6

4

Q4: 1. $(A \cap B)' - C$

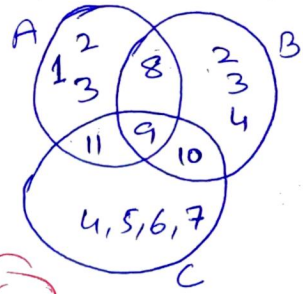
$$U = \{1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11\}$$

$$A = \{1, 2, 3, 8, 9, 11\}$$

$$B = \{2, 3, 4, 8, 9, 10\}$$

$$C = \{4, 5, 6, 7, 9, 10, 11\}$$

7/10



$$A \cap B = \{2, 3, 8, 9\}$$

$$(A \cap B)' = U - (A \cap B) = \{1, 4, 5, 6, 7, 10, 11\}$$

$$(A \cap B)' - C = \{1\}$$

5

2. $A' \cup (B \cup C)$

$$A' = U - A = \{4, 5, 6, 7, 10\}$$

$$B \cup C = \{2, 3, 4, 5, 6, 7, 8, 9, 10, 11\}$$

$$B \cup C = \{2, 3, 4, 5, 6, 7, 8, 9, 10, 11\}$$

$$A' \cup (B \cup C) = \{4, 5, 6, 7, 9, 10\}$$

Q5:

$$A - B = A \cap B'$$

let

~~$x \in A \text{ and } x \notin B$~~

definition of:

$x \in A - B$

$\Rightarrow x \in A \text{ and } x \notin B$

(set difference)

$\Rightarrow x \in A \text{ and } x \in B'$

(complement)

$\Rightarrow x \in A \cap B'$

(intersection)

But x is arbitrary element of $A - B$

$\therefore A - B \subseteq A \cap B' \rightarrow (i)$

Conversely

let

$y \in A \cap B'$

definition of:

~~$y \in A \text{ and } y \in B'$~~

(intersection)

~~$y \in A \text{ and } y \notin B$~~

(complement)

$y \in A - B$

(set difference)

but y is arbitrary element of $A \cap B'$

~~$\therefore A \cap B' \subseteq A - B \rightarrow (ii)$~~

From (i) & (ii) we get

$A - B = A \cap B'$

8/8

10/12

Q6:

1. $2 \in \{1, 2, 3\}$ True ✓

4. $\{2\} \subseteq \{1, 2, 3\}$ True ✓

2. $\{2\} \in \{1, 2, 3\}$ False ✓

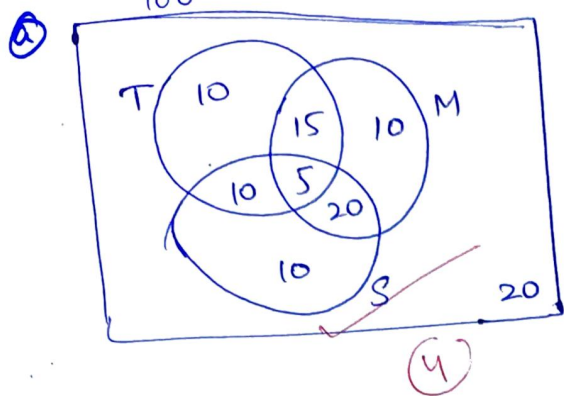
5. $\{2\} \subseteq \{\{1\}, \{2\}\}$ ^{True} False ✓

3. $2 \subseteq \{1, 2, 3\}$ False ✓

6. $\{2\} \in \{\{1\}, \{2\}\}$ True

(5)

Q7:



- (b) ~~10+20~~ 20
- (c) $10 + 15 + 10 + 10 + 5 + 20 = 70$ (2)
- (d) $10 + 15 + 10 = 35$ (2)
- (e) $10 + 20 + 15 = 45$ (2)
- Calculation?

11/12

Q8:

$$U = \{1, 2, 3, 4, 5, 6, 7, 8, 9, 10\}$$

(a) $\{2, 3, 4, 7, 8, 9, \}$

8/8

Bit string = 0, 1, 1, 1, 0, 0, 1, 1, 1, 0

0111001110

(4)

$$U = \{1, 2, 3, 4, 5, 6, 7, 8, 9, 10\}$$

(b) $\{3, 4, 5\}$

Bit string = 0, 0, 1, 1, 1, 0, 0, 0, 0, 0

0011100000

(4)